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Midazolam Dosing During CRRT: A Combined Ex Vivo and Physiologically-Based Pharmacokinetic Approach

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ABSTRACT

Children supported with continuous renal replacement therapy have high mortality rates ranging from 30% to 70%. The cause of this high mortality is multifactorial and includes ineffective drug dosing and altered drug pharmacokinetics. Changes in drug exposure can result from (1) underlying disease; and (2) direct drug interaction and/or clearance by the circuit. The extent to which these effects interact and modify drug pharmacokinetics is currently unknown for most drugs. As a result, there is little guidance on drug dosing in this population. We have developed a combined ex vivo and physiologically-based pharmacokinetic modeling approach to mechanistically model midazolam concentrations in pediatric patients on continuous renal replacement therapy. Published models were scaled to children and modified to incorporate patient-specific physiological changes due to critical illness including residual renal function, hematocrit, albumin levels, and inflammation-induced downregulation of drug metabolizing enzymes. A compartment representing the dialysis circuit was created, parameterized using ex vivo data, and added to the full child physiological based pharmacokinetic model. The child model was then used to predict drug exposure in children undergoing continuous renal replacement therapy. Observed data from an ongoing opportunistic pharmacokinetic study were used to validate the models. The midazolam continuous renal replacement therapy models accurately represented drug concentrations and were used to predict optimal drug dosing. Model informed dosing for all age groups fell within the standard dosing range suggesting no changes are needed during continuous renal replacement therapy.

1 | Introduction

Continuous renal replacement therapy (CRRT) is a type of hemodialysis typically used in critically ill patients. During CRRT, blood is pumped from the patient, through a circuit where it is filtered and before being returned to the patient. This process removes toxins and maintains fluid, electrolyte, and metabolic

balance during periods of acute or chronic kidney failure. Despite correction of major metabolic alterations, mortality rates are high, averaging between 30% and 70% [1, 2]. Mortality is likely influenced by alterations in drug exposure due to both patient-specific factors (e.g., age, underlying medical conditions, and organ failure) and CRRT circuit-related factors (e.g., adsorption to circuit components and drug clearance by the hemodialysis filter).

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Study Highlights

- What is the current knowledge on the topic?
 - Midazolam is a commonly used medication for sedation and analgesia in children supported by CRRT. Dialysis can affect the PK of drugs, resulting in altered drug exposure. Currently, there are no dosing guidelines for midazolam specific for this population.
- What questions did this study address?
 - This study provides pediatric dosing guidance for midazolam during CRRT that accounts for drug-CRRT circuit interactions and critical illness.
- What does this study add to our knowledge?
 - This study provides a novel method for determining appropriate midazolam dosing for children supported by CRRT.
- How might this change drug discovery, development, and/or therapeutics?
 - This study combined ex vivo data with PBPK modeling to develop a CRRT-PBPK model capable of accounting for drug extraction by the CRRT circuit and physiologic alterations due to critical illness. The model accurately simulates drug concentrations in children supported by CRRT. This modeling approach can be used to determine dosing guidance for other medications commonly used during CRRT.

Traditionally, drug dosing in children on CRRT is determined through conventional pharmacokinetic (PK) studies, which characterize drug disposition to establish safe and effective dosing in specific patient populations. Applying traditional PK methods in children supported with CRRT presents significant challenges. Drug disposition in this population varies based on the specific drug, circuit components, and patient age. This necessitates a PK study for each drug, circuit manufacturer, and age group [3–5]. Having sufficient patient numbers to comprehensively characterize PK across all variables is not feasible. Therefore, optimal dosing for most drugs in this population remains unknown [2].

While drug adsorption and clearance by the CRRT circuit are well-recognized contributors to PK variability, predicting drug-circuit interactions remains difficult. Ex vivo studies have been used to both identify the role of circuit components in drug loss and estimate PK changes during CRRT [6–13]. However, these studies are conducted in a closed-loop system lacking metabolic or elimination processes. Therefore, observed drug loss is attributed solely to circuit adsorption, filter clearance, or drug degradation. Consequently, ex vivo experiments provide valuable insights into circuit-related drug loss; however, they do not account for patient physiology, thus limiting their ability to predict in vivo drug exposure.

Physiologically-based pharmacokinetic (PBPK) models offer a mechanistic approach to integrating both patient- and circuit-related factors that affect drug disposition in critically ill

children undergoing CRRT. PBPK models are mathematical tools that integrate drug-specific and population-specific information to characterize the effect of different factors (e.g., age, critical illness, CRRT modality) on drug exposure [14, 15]. These models use physiologically-relevant compartments representing organs connected by blood flow. Differential equations within each compartment characterize changes in drug disposition over time. To model drug exposure in children on CRRT, a pediatric PBPK model can be developed using standard approaches [15–19]. A CRRT virtual compartment can then be added to this model and parameterized using ex vivo data. Incorporation of age-related physiological factors that influence PK in children enables the development of a mechanistic PBPK model capable of estimating drug exposure in children of all ages who are supported by CRRT [15, 19–20].

In this study, we developed a midazolam CRRT-PBPK model to simulate midazolam concentrations in critically ill children undergoing treatment with various CRRT modalities. Using these simulations, we have developed model-informed midazolam dosing recommendations, offering a novel approach to optimizing drug therapy in this vulnerable population.

2 | Methods

2.1 | Software

PBPK models were created using PK-Sim and MoBi version 10 (Open Systems Pharmacology Suite, www.open-systems-pharmacology.com). Analyses and visualizations were completed with R Version 4.2 using additional packages: table 1, drc, tidyverse, ggpubr [21, 22].

2.2 | General Workflow

A 17-compartment population PBPK model was first developed in adults to validate the model's structure. The adult intravenous (IV) midazolam PBPK model was adapted from two published midazolam PBPK models developed in PK-Sim [23, 24]. Published drug-specific information (physicochemical properties, e.g., lipophilicity, molecular weight, clearance pathways) and system-specific information (e.g., organ size, blood flows) were used to parameterize the model [15, 19–20]. The adult PBPK model was then scaled to children by incorporating age-dependencies in physiological parameters. Mechanistic drug clearance (CL) was used to scale adult plasma CL to children using (1) adult CL, (2) the proportion of total CL attributed to each clearance process (e.g., renal vs. hepatic), (3) the fraction of unbound drug and major protein responsible for binding, and (4) age [15, 19]. The child model was evaluated and optimized using published PK data [25–27]. A compartment representing the CRRT circuit was created, parameterized using ex vivo data, and added to the validated child PBPK model. The midazolam child CRRT-PBPK model was used to predict drug exposure in three children of varying ages undergoing different CRRT modalities. Observed data from an ongoing opportunistic PK study of drugs in children on CRRT were used to evaluate CRRT-PBPK models. Average fold error (AFE), absolute average fold error (AAFE) and number of observations within

a 90% prediction interval (5%–95%) were used to assess model bias and precision.

2.3 | Adult PBPK Model

We developed an adult PBPK model by adapting two published IV midazolam PBPK models developed in PK-Sim [24, 25]. Drug-specific data including physicochemical properties (e.g., lipophilicity, molecular weight, CL pathways) were integrated into the model (Table 1). Using parameters from the two previously published PBPK models, we created a third midazolam model. To optimize this model, we simulated IV midazolam exposure for a population of healthy adults and compared the simulated exposures with observed concentrations. For these simulations, virtual populations of healthy adults (European or American) were generated in PK Sim (Table S1) with system-specific parameters (e.g., organ size, blood flows) provided by the software

[15, 19–20]. For model evaluation, we divided the healthy adult PK datasets from the literature into two groups: (1) development datasets, and (2) validation datasets. The development datasets were used to optimize the model for IV midazolam formulations [32, 33]. The validation datasets were used to test the accuracy of model-generated predictions of drug exposure [34, 35]. A Monte Carlo-based parameter identification algorithm was used within PK-Sim to optimize key drug properties including logP, and fraction unbound. Once the optimized healthy adult models met acceptance criteria (see Section 2.8), it was then scaled to children.

2.4 | Child PBPK Model

To scale the model to children, drug-specific parameters remained unchanged while physiological parameters were adjusted based on age-dependent changes. These age-dependent parameters

TABLE 1 | Midazolam physicochemical properties and PBPK base model parameters.

Parameter	Optimized	Literature	Source
Physicochemical properties			
Lipophilicity (LogP)	2.99	2.76, 3.13	Salerno et al. (2021) [22]; Hanke et al. (2018) [23]
Plasma binding protein		Albumin	Gertz et al. (2010) [28]
Fraction unbound (%)	2.0	3.0, 1.6	Salerno et al. (2021) [22]; Hanke et al. (2018) [23]
Molecular weight (g/mol)		325.77	Drugbank. Drugbank 5.0, 2018
Effective molecular weight		286.78	Calculated, PK-Sim software
Compound type		Base	Drugbank. Drugbank 5.0, 2018
pKa		6.57	Drugbank. Drugbank 5.0, 2018
Charge		Neutral	Drugbank. Drugbank 5.0, 2018
Solubility (mg/mL)		0.05	Heikkinen et al. (2012) [29]
Solubility reference pH		6.5	Heikkinen et al. (2012) [29]
Absorption			
Specific intestinal permeability (cm/min)	1.55E-4	2.00E-5	Hanke et al. (2018) [23]
Specific organ permeability (cm/min)		0.05	PK-Sim software
Distribution			
Partition coefficients		Rodgers and Rowland	PK-Sim software
Cellular permeabilities		PK-Sim Standard	PK-Sim software
Metabolism and elimination			
GFR		1	Hanke et al. (2018) [23]
CYP3A4			
Km (μM)		1.88	Xiao et al. (2019) [30]
V _{max} (L/min)		6.12	Xiao et al. (2019) [30]
CYP3A7			
Km (M)		11.4	Williams et al. (2002) [31]
V _{max} (L/min)		4.0	Williams et al. (2002) [31]

included body weight, height, organ weight, organ maturation, blood flow, protein binding, and drug CL [15, 19–20]. Adult drug CL pathways were modified as a function of child age [15, 19]. The ontogeny of the metabolic enzyme cytochrome P450 (CYP) 3A4 was accounted for within PK-Sim using age-based ontogeny factors [36]. To account for drug CL in neonates and preterm infants, CYP3A7 metabolism was included in the model using data from a published in vitro study (Table 1) [36, 37].

2.5 | CRRT Compartment

To account for the impact of CRRT on drug exposure, previously published ex vivo experimental data were used to develop and parameterize a CRRT compartment in MoBi [13]. The CRRT compartment consisted of three sub-compartments representing the dialysis machine's (1) Filter; (2) Filter dialysate; and (3) Dialysate bag. The filter compartment was further divided into plasma and red blood cell sub-compartments (Figure 1). In addition, a virtual compartment representing a reservoir of blood was built and divided into plasma and red blood cell sub-compartments to represent the ex vivo experimental setup (Figure 1B). The CRRT compartment was then connected to the reservoir through the blood and plasma sub-compartments to represent the ex vivo experimental set up (i.e., ex vivo CRRT model). Volumes and flow rates from ex vivo experiments were included in the ex vivo CRRT model. A reservoir volume of 600 mL and hematocrit of 0.30 were used, representing the average values obtained across three replicates of ex vivo experiments. Parameters for the HF1000 hemofilter (Baxter Healthcare, Deerfield, IL) were added to the CRRT

compartment as follows: dialysate volume 59.1 mL, filter volume 165 mL, surface area 1.2 m², and membrane thickness 50 µm. The volume of dialysate was equivalent to HF1000 filter prime volume. The filter volume represented both the HF1000 filter and circuit prime volumes to account for the additional volume of the circuit. The CRRT prescription used for ex vivo experiments was as follows: blood flow rate, 80 mL/min; dialysate flow rate, 400 mL/min; pre-blood pump, 300 mL/h; replacement fluids, 100 mL/h; and patient fluid removal, 0 mL/h. Additional flow rates equations were derived from the PrisMax Operator's Manual (S1) [38].

Ex vivo concentration data were used to evaluate and optimize the CRRT compartment. Model simulations were run using a midazolam dose of 0.15 mg/L to simulate the exposure achieved during the original ex vivo experiments. Parameter optimization (Monte Carlo simulation in PK-Sim) was used to adjust the elimination rate constant and membrane diffusion coefficients until model simulations matched ex vivo concentration data (Figure 1A).

After optimization, the CRRT compartment was connected to the midazolam child PBPK model through the venous blood compartment (Figure 1B). The model was then evaluated using observed concentration data from children on midazolam and CRRT (Figure 1B).

2.6 | Opportunistic PK Study

Patient blood and effluent samples were collected as part of a prospective opportunistic PK study in children supported with CRRT

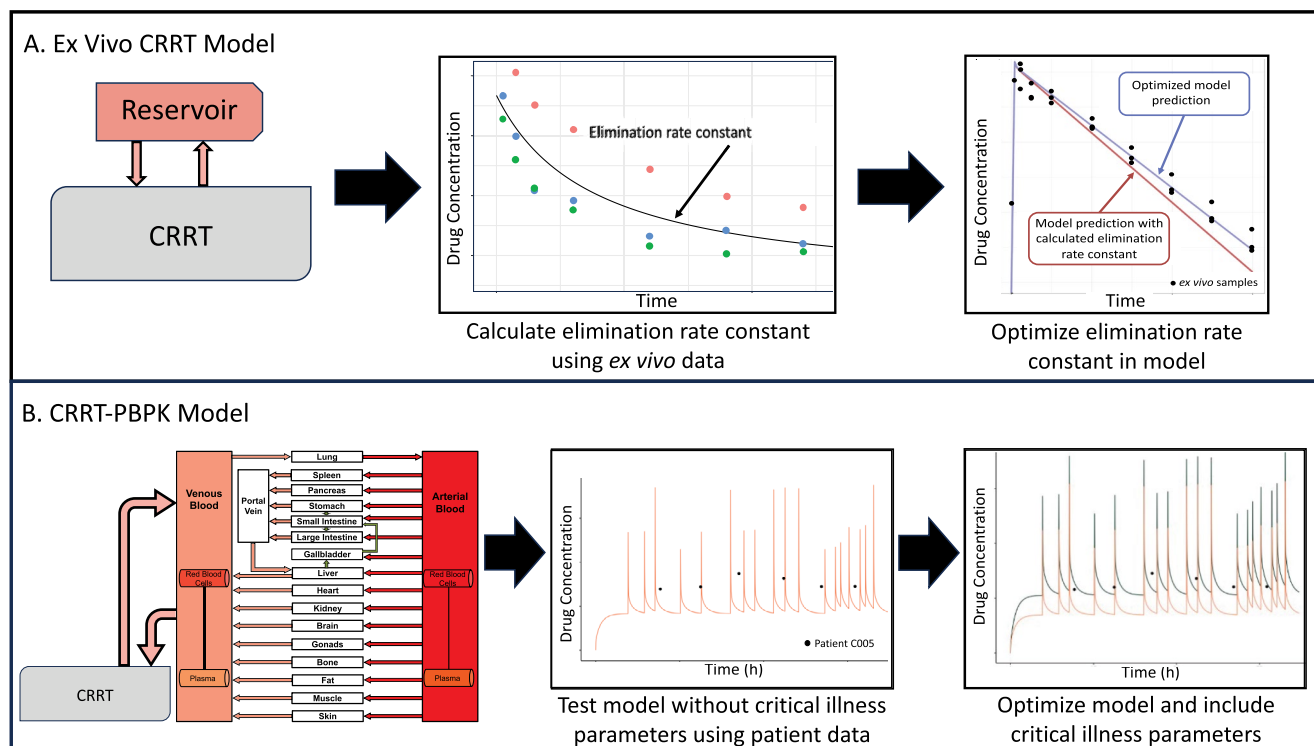


FIGURE 1 | CRRT compartment development. Ex vivo data were used to calculate an elimination rate constant (A). A CRRT compartment with a reservoir was built and simulations were run using ex vivo circuit dosing. The elimination rate constant was optimized to ex vivo concentration data. Once optimized, the reservoir was removed and the CRRT compartment connected to a full PBPK model (B). Model simulations were run for patients and observed data were compared to model predictions.

at Primary Children's Hospital, Salt Lake City, Utah (Opportunistic Pharmacokinetic Study of Drugs Administered per Standard of Care to Children on Extracorporeal Life Support, IRB 00138310). Inclusion criteria included: (1) age between 0 and 18 years; (2) undergoing CRRT; and (3) administered a study drug per standard of care. Informed consent was obtained from a legal guardian and assent was obtained from the child if age was above 12 years and not sedated. Approximately 0.5 mL of blood and 1 mL of effluent were collected opportunistically at the time of scheduled labs. The exact date, time, route, site of administration, and formulation of study drugs; clinical data (e.g., age, sex, race, gestational age, drug indication, laboratory tests, medical history, physical exam, and concomitant medications); and adverse events after the first dose and for 3 days after the last PK sample collection were recorded.

2.7 | CRRT-PBPK Model

For model evaluation, a virtual individual was created in PK-Sim to represent each patient for model evaluation (Table S2). Patient-specific parameters included midazolam dosing regimen, CRRT prescription, and demographics. After initial model evaluation, critical illness factors (e.g., albumin, hematocrit, GFR prior to CRRT initiation, and reduced CYP activity) were added in MoBi to improve model performance. We anticipated that changes in albumin would impact the protein binding of midazolam. To account for this, the plasma protein scaling factor (PPSF) was adjusted based on both Wu et al. and Johnson et al. [38, 39]. PPSF was calculated using Equation (1), which was obtained from Wu et al. (2022) and adapted from the original equation published by Johnson et al. (2010). Here, we used the adapted equation from Wu et al. to calculate our PPSF. The adjusted, calculated value was then added to the model.

Equation 1: PPSF where f_u is the simulated fraction unbound, $f_{u,ref}$ is the reference fraction unbound from age-matched individuals, and $Albumin_f$ is the fraction of albumin in the plasma compared to healthy, age-matched individuals.

$$PPSF = 1 / (f_u + (1 - f_{u,ref}) \times Albumin_f) \quad (1)$$

To account for reductions in CYPs due to inflammation, CL was calculated with and without inflammation using the equation from Brussee et al. [40].

Equation 2: Metabolic CL accounting for inflammation for an individual patient. WT is weight, and CRP is C-reactive protein.

$$CL = 1.29 \text{ L/h} \times (WT/5)^{1.02} \times (CRP/32)^{-0.312} \quad (2)$$

A value of 1.29 L/h represents the presence of a single organ dysfunction (e.g., kidney for this study) and was used for all patients. This value is adjusted based on the number of failing organs with 1.29, 0.96, 0.83 and 0.67 L/h representing 1, 2, 3, and 4 organs, respectively. The type of organ is not relevant for this calculation of total CL [39]. The CL was calculated for each patient with and without inflammation. For example, in a patient weighing 10 kg and a normal CRP of 10 mg/L, the calculated CL would be 3.78 L/h. If CRP was then elevated to 50 mg/L, the CL would be calculated as 2.31 L/h. This would represent a 64% decrease in CL due to inflammation. The V_{max}

would then be adjusted until a matching 64% reduction in CL was achieved.

Filter parameters were also adjusted based on the type of filter used for each patient (Table S3). Finally, using periods of time with stable flow rates, average flow rates were determined across the simulation interval. Simulations represented continuous periods of dialysis and did not include circuit changes. After model evaluation using patient samples, population CRRT-PBPK simulations were performed to determine optimal dosing.

2.8 | PBPK Model Acceptance Criteria

Population model bias for mean observed data was evaluated by calculating the AFE between predicted and observed concentrations [22, 36]:

Equation 3: Average fold Error

$$AFE = 10^{[1/N \sum \log(\text{predicted}/\text{observed})]} \quad (3)$$

AFE values below and above 1 indicate under- or over-prediction, respectively. Overall model accuracy was determined using the absolute average fold error [22]:

Equation 4: Absolute average fold error.

$$AAFE = 10^{[1/N \sum |\log(\text{predicted}/\text{observed})|]} \quad (4)$$

Model acceptance was met when $AFE = 0.7\text{--}1.3$ and $AAFE \leq 3$ [22]. Population predictability was evaluated using the number of observations that fell within a 90% prediction interval (5%–95%). Models were considered acceptable if > 90% of observed concentrations fell within the 90% prediction interval.

2.9 | CRRT-PBPK Model Informed Dosing

To predict dosing across the age spectrum and account for age-specific changes in CRRT prescription, we created five age-stratified virtual populations: (1) neonatal (0–29 days); (2) infant (1 month to < 2 years); (3) child (2 years to < 6 years); (4) school age (6 years to < 12 years); adolescent (12 to 18 years) (Table S4). Each virtual population consisted of 1000 simulated individuals. CRRT prescriptions were informed by typical clinical practice at Primary Children's Hospital in Salt Lake City, Utah (Table S5). Neonatal blood flow rates were kept at 20 mL/min and infants at 30 mL/min due to filter limitations. Individual model simulations were performed in MoBi and exported to PK-Sim for population models. Critical illness parameters for hematocrit, albumin, GFR prior to CRRT initiation, and CYP downregulation were averaged across study participants and included in all virtual populations (Table S4). Specifically, the following parameters were included across all pediatric populations: a GFR reduction of 75%, hematocrit reduction of 28%, albumin reduction of 37%, and a reduction in CYP metabolism of 26% compared to healthy age-matched children. To ensure critical illness parameters were appropriate, midazolam CL in virtual critical illness populations without kidney dysfunction was compared to published studies. GFR was adjusted to

reflect values found within the CRRT population. Within each age group the CRRT-PBPK model was evaluated across 30 midazolam continuous infusion doses ranging between 0.008 to 0.30 mg/kg/h. Drug doses were considered optimal if midazolam concentrations for 90% of virtual individuals were within the therapeutic range of 200–1000 ng/mL at steady state [41].

2.10 | Midazolam Assay

Midazolam patient plasma and effluent drug concentrations were measured at OpAns Laboratory (Durham, NC) using high-performance liquid chromatography-mass spectrometry with a lower limit of quantification (LLOQ) of 100 pg/mL. Standard curve acceptance was achieved if accuracy was within 85%–115% for at least 75% of standards. A minimum coefficient of determination (r^2) of at least 0.96 within the calibration range of 100 to 200,000 pg/mL was required. The r^2 had a mean of 0.997 (standard deviation of 0.0028). Accuracy within and between runs ranged from 89.2%–116.5% and 94.2%–106.7%, respectively. Within run precision ranged from 2.26% to 8.48% and between run precision ranged from 3.47% to 10.48%.

3 | Results

3.1 | Model Evaluation: Adult

Midazolam adult PBPK model parameters can be found in Table 1. In healthy adults, the average literature values for area under the curve (AUC), CL, terminal half-life ($T_{1/2}$), and volume of distribution (V_d) were reported as 0.015 mg×min/mL, 410 mL/min, 2.52 h, and 0.96 L/kg respectively [42–45]. Model predictions agreed with literature values (Figure 2A, Figure S1). The final adult midazolam model met acceptance criteria with an AFE of 0.92 and AAFE of 1.52.

3.2 | Model Evaluation: Child

The child model showed good agreement with observed data from the literature (Figure 2B, Table S6) with an AFE of 1.23, AAFE of 2.18, and 94% of the observed child data falling within the 90% population confidence interval [39, 40]. The preterm

model including CYP3A7 showed good agreement with observed data from preterm infants (Figure 2C) [40]. The preterm model AFE (1.53) and AAFE (1.53) were acceptable, and 100% of the preterm observed data were within the 90% confidence interval.

3.3 | CRRT Compartment Parameterization

Using ex vivo experimental data, a Michaelis–Menten elimination rate constant was calculated in *R* and further optimized in MoBi, yielding a final K_m of 1100 pmol/mL and V_{max} of 0.002 L/min. The final optimized elimination rate constant was used for all subsequent models.

3.4 | Opportunistic PK Study

A total of 3 patients aged 12, 22, and 72 months were enrolled in the prospective opportunistic PK study and used to evaluate the CRRT-PBPK model. The patients represented two modalities of CRRT: continuous venovenous hemodialysis (CVVHD) and continuous venovenous hemodiafiltration (CVVHDF). Midazolam infusions ranged from 0.01 to 0.1 mg/kg/h IV, with boluses of the same concentrations given over the course of the simulations (Figure 4A). A mean of 9 (range: 3–13) paired blood and effluent samples were collected for each study participant over 14 days. One patient was treated with plasmapheresis during the study period. In this patient, model simulations were started 5 h after plasmapheresis and 4 h after the start of CRRT, post-treatment. Patient samples were collected 6 and 24 h after the end of plasmapheresis.

3.5 | Model Evaluation: CRRT-PBPK

An initial model simulation for an IV infusion of 0.1 mg/kg/h was run to test the effect of CRRT circuit clearance on simulated plasma concentrations (Figure 3). The addition of the CRRT compartment resulted in a reduced plasma concentration indicating drug loss due to clearance by the CRRT circuit. The initial CRRT-PBPK model underpredicted concentrations (Figure 4B). Adults and children on CRRT commonly experience low albumin, hematocrit, and inflammation-induced downregulation

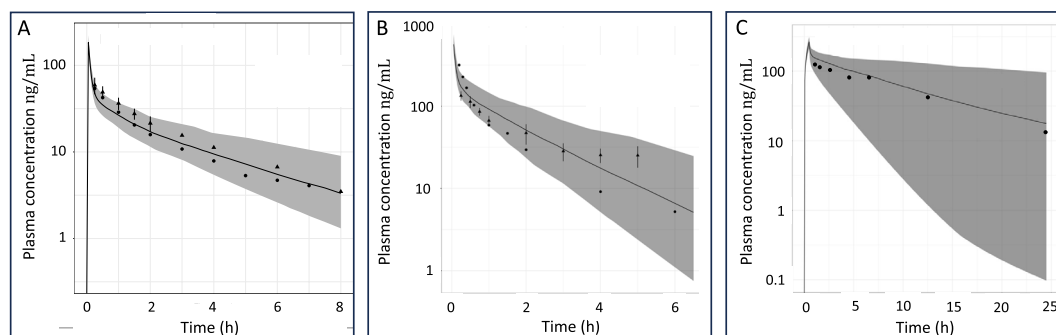


FIGURE 2 | Midazolam PBPK population model. Adult midazolam PBPK population model for a single midazolam IV dose of 0.05 mg/kg (A). Child midazolam PBPK population model for a single midazolam IV dose of 0.15 mg/kg (B). Preterm infant midazolam PBPK population model for a single midazolam IV dose of 0.1 mg/kg (C). The dark solid line is the model predicted median. Shaded areas are the 90% confidence interval (5%–95%). Black circles and triangles represent mean observed data from PK studies [24, 26, 33–34].

of CYPs. To improve model predictions, these parameters were adjusted for each patient simulation to reflect disease state (Figure 4C). Individual model simulations were in good agreement with all observed data except patient 3 (Figure 4, Figure S2). Final model performance was acceptable with all patient samples within the 90% confidence interval (Figure 5). In age groups where more than one filter type was appropriate (child, school age and adolescent), models were run for each filter type (data not shown). No differences were found in drug concentration between filters within each age group. Additional simulations were run with and without replacement fluids and pre-blood pump flow to simulate different modalities of CRRT. No differences were found in simulated drug concentrations

between the different CRRT modalities. Model calculated effluent doses for individual patients were similar to values recorded in patient records.

3.6 | Dosing Guidance

The midazolam CRRT-PBPK model was used to simulate different doses of midazolam continuous infusions for critically ill pediatric patients. Across all age groups, midazolam clearances of virtual critical illness populations without kidney dysfunction were 3.6–11.3 mL/min/kg compared to 0.5–13 mL/min/kg in the literature [41–44]. Model predicted optimal dosing for

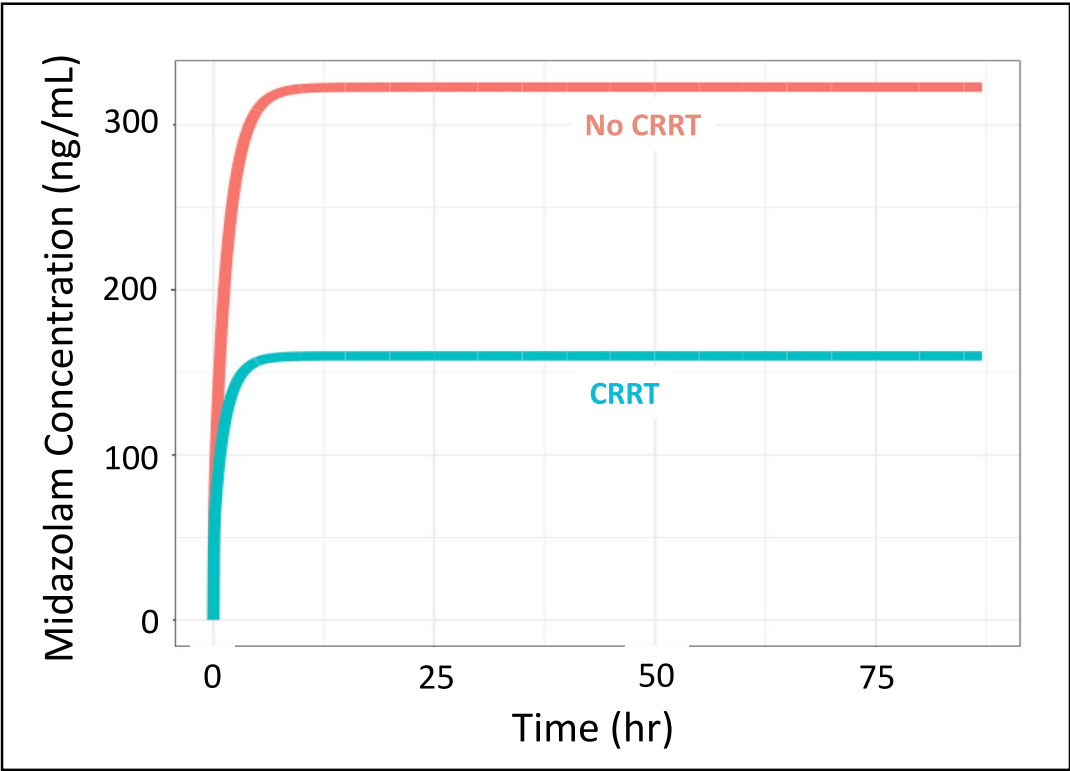


FIGURE 3 | Model simulations with and without the CRRT compartment. Model-predicted concentrations for 0.1 mg/kg/h infusion of midazolam for an individual with (solid blue line) and without (solid pink line) CRRT compartment. Patient parameters included age, GFR prior to the start of CRRT, and CRRT prescription. No additional critical illness parameters were included in this simulation.

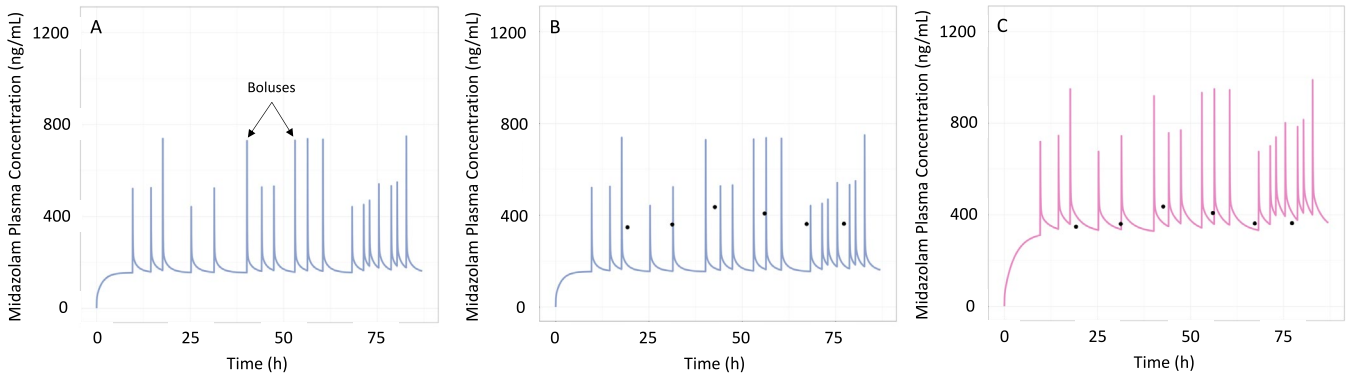


FIGURE 4 | Individual model predicted midazolam compared to observed data. Model simulations of an individual patient (patient 1) with an infusion of 0.1 mg/kg/h and boluses of 2.2 mg (A). The model simulation was compared to patient plasma concentration (solid black circles) without (B) and with (C) critical illness parameters. Solid lines represent simulated model concentration predictions with (blue) and without (pink) critical illness.

midazolam was 0.04 mg/kg/h for neonates and 0.1 mg/kg/h for all other age groups (Figure 6).

4 | Discussion

Dosing guidance for children on CRRT is limited and the use of traditional methods to provide dosing guidance presents significant challenges. In this study, a combined ex vivo and PBPK modeling approach was used to provide a feasible method for predicting optimal midazolam dosing in critically ill children supported with CRRT. Initially, an adult model was developed and verified before scaling to children. Both the adult and child models performed well and served as the base structure for subsequent CRRT-PBPK models. To account for the impact of CRRT, a CRRT compartment was developed and parametrized using ex vivo experimental data. Our ex vivo studies demonstrated that midazolam primarily absorbed to CRRT circuit components but was minimally cleared through the hemofilter [12]. This was due to the high lipophilicity and albumin binding properties of midazolam. The CRRT compartment

was integrated into the child PBPK model and residual kidney function was adjusted to reflect decreased renal function. We used previously published literature to compare and confirm that the parameters for critical illness we used were appropriate. The model was then evaluated using plasma concentration data from children supported with CRRT. The model initially under-predicted patient concentrations. To improve accuracy, parameters reflecting critical illness, including reduced albumin levels, lower hematocrit, and decreased liver enzyme activity due to inflammation, were added. The inclusion of these critical illness parameters greatly improved the model performance.

Critical illness can significantly impact the PK of drugs with key alterations driven by inflammation, anemia, and hypoalbuminemia [45–47]. The effect of critical illness on PK is determined by the severity of disease and drug-specific properties. The high protein binding and CYP-mediated liver metabolism of midazolam make this drug susceptible to changes in albumin levels and CYP activity. Reduced production and increased loss of albumin associated with both critical illness and kidney failure can result in a high proportion of unbound drug [45–49]. This

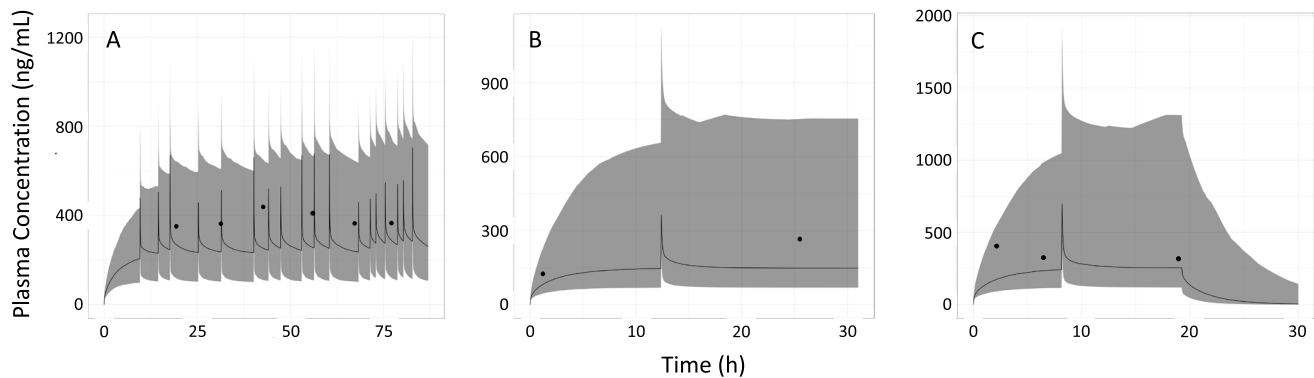


FIGURE 5 | Population CRRT-PBPK model evaluation with observed data. Midazolam continuous infusions ranged from 0.06 mg/kg/h for patient 2 (B) to 0.1 mg/kg/h for patients 1 and 3 (A, C) with additional boluses at the same concentration (e.g., 0.1 mg/kg/h infusion with 0.1 mg/kg boluses). Solid lines represent the model-predicted median and shaded areas are the 90% confidence interval (5%–95%). Black circles represent observed data pediatric patients supported with CRRT.

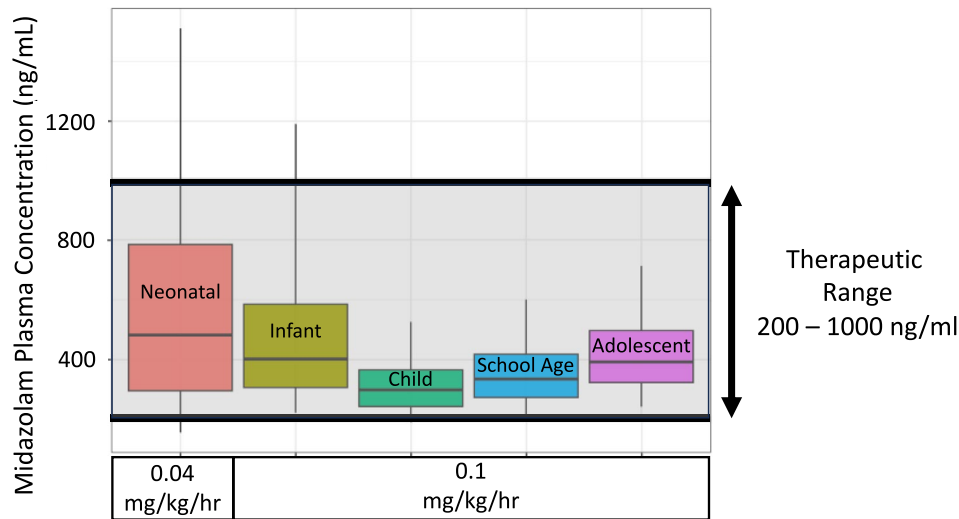


FIGURE 6 | Model-informed dosing guidance. Model-informed dosing guidance for midazolam sedation with a continuous IV infusion in children supported with CRRT. The gray shaded area represents the therapeutic range for midazolam of 200–1000 ng/mL. Boxes represent interquartile ranges for each virtual population by age group. Solid lines within the boxes represent the median and whiskers represent the 90% prediction interval.

higher unbound drug concentration could increase target tissue drug concentrations. However, CRRT can clear unbound drug and thus mitigate, to some extent, increases in drug concentrations. To recapitulate the ratio of bound and unbound fractions of midazolam found within our target population and accurately account for CRRT circuit CL of unbound drug, we used a lower albumin concentration in our ex vivo circuits. CL of midazolam can also be altered by inflammation-induced downregulation of CYP-mediated metabolism [40, 50–52]. Decreases in CL have been directly associated with both the severity of inflammation and degree of organ dysfunction present in critically ill children [40, 53]. The impact of critical illness on midazolam PK can result in high drug concentration variability within these pediatric populations [53]. Critical illness parameters were included in the CRRT-PBPK model to better represent the patient population. This increased the accuracy and precision of model predictions. All individual CRRT-PBPK models met acceptance excluding patient 3 who received fluconazole, a moderate CYP3A4 inhibitor, during the simulation period. The drug–drug interaction between fluconazole and midazolam may account for the underprediction of midazolam concentration.

The final CRRT-PBPK model incorporating CRRT prescription and critical illness parameters was used to predict optimal dosing within the therapeutic range of 200–1000 ng/mL [39, 41]. Standard dosing guidelines recommend 0.015–0.06 mg/kg/h for neonates and 0.05–0.12 mg/kg/h for all other pediatric age groups [49]. The CRRT-PBPK model informed dose of 0.04 mg/kg/h for neonates and 0.1 mg/kg/h for all other pediatric age groups is aligned with standard dosing recommendations. These findings suggest that no dosing adjustment for midazolam is needed in children supported with CRRT, consistent with results from ex vivo and patient studies [12, 49]. Importantly, midazolam is a highly protein bound drug with low filter clearance. Thus, the clearance of midazolam by the CRRT circuit may be flow dependent. Additional studies are needed to determine the significance of flow rates in filter clearance and midazolam dosing.

We chose IV midazolam as our drug of interest, as midazolam is a commonly used sedative in pediatric and neonatal critical care units. Infants and children on CRRT often receive IV midazolam as a sedative and anxiolytic. Although midazolam is titrated to effect, there exists conflicting data on midazolam clearance by CRRT. Currently, no optimal dosing guidelines exist for pediatric IV midazolam use in CRRT. A well-known side effect of IV midazolam cumulative use is drug-induced delirium. Delirium in children causes confusion and has the potential to increase the length of stay. Thus, it is imperative to determine optimal dosing of midazolam to achieve and maintain therapeutic effects while minimizing the risk of delirium in children.

There are several limitations to this study. Ex vivo data used for model development were determined from only one modality of CRRT (CVVHDF) and one filter type (HF1000). Filter clearance of drugs likely differs by hemodialysis modality. Nevertheless, the CRRT-PBPK model was able to accurately simulate midazolam concentrations for CVVHD and CVVHDF modalities. Flow rates can vary based on the age and size of the patient. Age-appropriate flow rates were

incorporated into the model using representative individuals for each age group. While the CRRT-PBPK models accurately simulated midazolam concentrations when compared to patient data, the analysis was limited by the small number of patients and samples. Additional patient data are needed to fully validate the CRRT-PBPK model and further investigate the effect of flow rate on midazolam CL. In addition, we did not incorporate clearance contributions from UGT. As the contribution of UGT to midazolam metabolism primarily occurs via secondary metabolites, here we focused only on midazolam concentrations. Despite these limitations, the CRRT-PBPK model performed well and accurately predicted drug concentrations in children supported with CRRT.

The midazolam CRRT-PBPK model accurately simulated drug concentrations in critically ill children supported with different modalities of CRRT. Model-informed dosing was within standard dosing ranges and suggests no dosing changes are needed for midazolam in the setting of CRRT. The CRRT compartment can be easily adapted to other drugs using ex vivo experiments to determine drug clearance by the circuit. This modeling approach provides a feasible method to guide dosing in this complex population.

Author Contributions

A.M.M. and K.M.W. wrote manuscript. A.M.M., A.H., J.P.H., D.J.G., A.E., and K.M.W. designed the research. A.M.M., J.P.H., D.J.G., C.I., A.W., R.H., A.C., and K.M.W. performed research. A.M.M., C.L.M., J.P.D., A.E., M.J.Y. and K.M.W. analyzed data.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** psp470240-sup-0001-Supinfo.zip.